

Are High-Interest Loans Predatory? Theory and Evidence from Payday Lending

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High-Interest Lending

High-interest lending is long questioned, leading to regulation.

- The effects of such regulation are yet poorly understood.

Payday loans: a typical form of high-interest lending

- Repaid on paydays
- Small, unsecured
- High-cost (annual rate 391%)
- Short-term (1-2 weeks)

Debates over Payday Loans

Critics: payday loans are predatory.

- They exploit peoples' *present focus* and *over-optimism*, trapping consumers in cycles of borrowing.

Proponents: payday loans serve a critical need.

- People are willing to pay high interests because they do need credit.

Key question: Are payday loan borrowers maximizing their utility, or affected by behavioral issues?

Key Behavioral Biases

Two important behavioral biases to capture:

1. **Naivete/over-optimism:** underestimate their likelihood of reborrowing and the costs of repayment
2. **Present focus/time-inconsistency:** self-control problems, overborrow to finance present consumption

Main Results

This paper uses an experiment to study two important behavioral biases of payday loan borrowers.

- The majority of borrowers correctly predict their likelihood of future borrowing.
- Borrowers perceive themselves to be time inconsistent.

It also studies and estimates a model of present focus and naivete.

- Borrowers are unlikely to be trapped in infinite debt.
- Banning payday loans reduces welfare, while limits on repeat borrowing might increase welfare.

Overall Structure of This Paper

Overall structure of this paper

1. Experiment design and implementation
2. Reduced-form identification of the behavioral biases
3. Theoretical results and structural estimation
4. Numerical simulation and implications

Experiment Design: Step 1

Key questions

Q1 (naivete): Do borrowers anticipate they'll reborrow?

Q2 (present focus): Are they willing to pay for self-control?

Step 1: Belief elicitation

1. Repeat borrowing likelihood (no incentive)
 - Report the probability of another payday loan in the next 8 weeks
 - Possible answers: 0%, 10%, ..., 90%, 100%
2. Introduction of "\$100 If You Are Debt-Free"
 - No more payday loan in the next 8 weeks → \$100 Visa cash card 12 weeks from now
3. Repeat borrowing likelihood (with incentive)
 - Report the probability of another payday loan in the next 8 weeks, given the no-borrowing incentive

Experiment Design: Step 2

Step 2: Valuation measurement via multiple price lists (MPLs)

1. First MPL: Valuation of the no-borrowing incentive

- Sequentially offer different Money For Sure, and bound the amount at which they are indifferent.

Survey Format

Earlier, you told us that you have a $[P]\%$ chance of getting another payday loan before [8 weeks from now] if you are selected for \$100 If You Are Debt-Free. In other words, you would have a $[100 - P]\%$ chance of being debt-free. So on average, \$100 If You Are Debt-Free would earn you $\$[100 - P]$.

Given that, which reward would you prefer?

- \$100 If You Are Debt-Free. This gives you extra motivation to stay debt-free.
- $\$[100 - P]$ For Sure. This gives you certainty and avoids pressure to stay debt-free.

Experiment Design: Step 2 & 3

Step 2: Valuation measurement via MPLs (Cont.)

2. Second MPL: Measurement of risk aversion

- Choose between Flip a Coin for \$100 and Money For Sure

Step 3: Randomization and incentive compatibility

- Assign participants randomly to three groups:
 - 44% Incentive group: Receive “\$100 If You Are Debt-Free”
 - 54% Control group: Receive no experimental reward
 - 2% MPL Group: Get their choice from one randomly selected MPL question
- Incentivize truth reporting

Experiment Implementation

Data

- 41 Lender's stores in Indiana
- 1205 borrowers with valid data
- January 7 – March 29, 2019 (2 consecutive weeks per store)



External validity

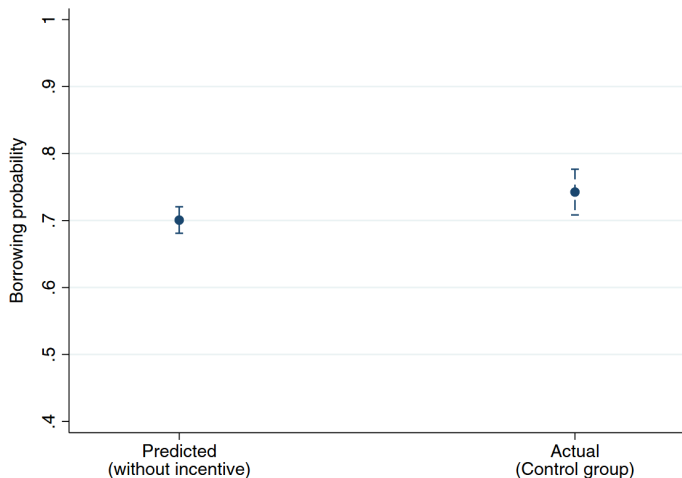
External validity: Nearly identical to 33,000+ national borrowers, no significant selection bias

TABLE 1
Descriptive statistics and external validity

	(1) Valid sample	(2) Customers on survey days	(3) 2017 loans nationwide
Loans in past six months	5.35 (2.94)		6.03 (4.22)
Annual income (\$000s)	34.0 (21.1)	31.8 (21.3)	28.9 (38.0)
Internal credit score	862 (122)	870 (193)	861 (125)
Pay cycle length (days)	16.0 (7.7)	18.1 (9.1)	17.3 (8.2)
Loan length (days)	17.3 (5.9)	18.7 (6.8)	17.5 (8.4)
Loan amount (\$)	373 (161)	359 (165)	366 (166)
<i>N</i>	1,205	13,191	33,194

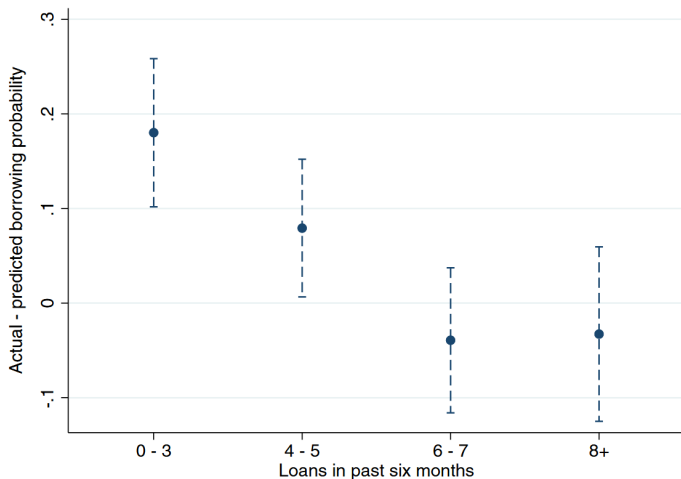
Q1: Do borrowers anticipate they'll reborrow?

The average borrower almost fully predicts repeat borrowing: 70% vs. 74%.



Q1: Do borrowers anticipate they'll reborrow?

Borrowers may learn from experience.



Q2: Are they willing to pay for self-control?

A risk-neutral, time-consistent person: \$40 for “\$100 Debt-Free”.

- Risk-averse borrowers: valuation should be less than \$40.

However, borrowers' average actual valuation is \$52, 30% more.

- Given their risk-aversion, this must reflect a demand for self-control.
- Consistent with qualitative survey responses
- Direct evidence of perceived time inconsistency

Q2: Are they willing to pay for self-control? (Cont.)



Identifying Sophistication/Naivete

Quasi-hyperbolic preferences (Laibson 1997; O'Donoghue and Rabin 1999)

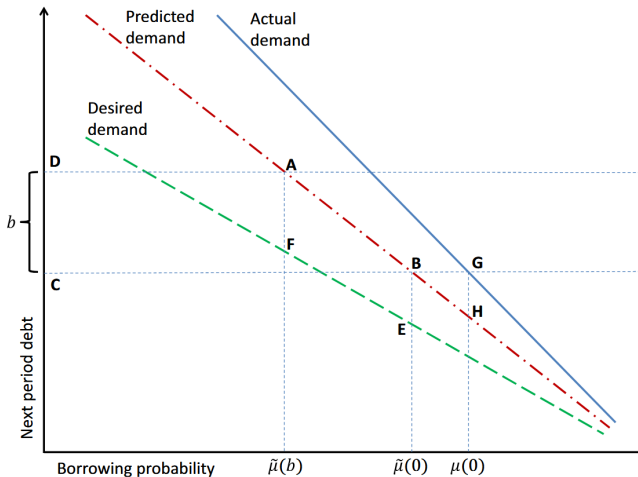
$$U_t = u_t + \beta\delta u_{t+1} + \beta\delta^2 u_{t+2} + \cdots ,$$

where

- $\delta \in (0, 1)$: long-run discount factor
- $\beta \in (0, 1)$: actual present focus
- $\tilde{\beta} \in (0, 1)$: perceived present focus (belief about β)

Sophistication/naivete: the ratio $\beta/\tilde{\beta}$ of actual to perceived present focus

Demand Gaps



Actual and Predicted demand differ: people mispredict future borrowing.

Predicted and Desired demand differ: people perceive themselves to be time-inconsistent.

Structural Estimation: Parameters

Parameters

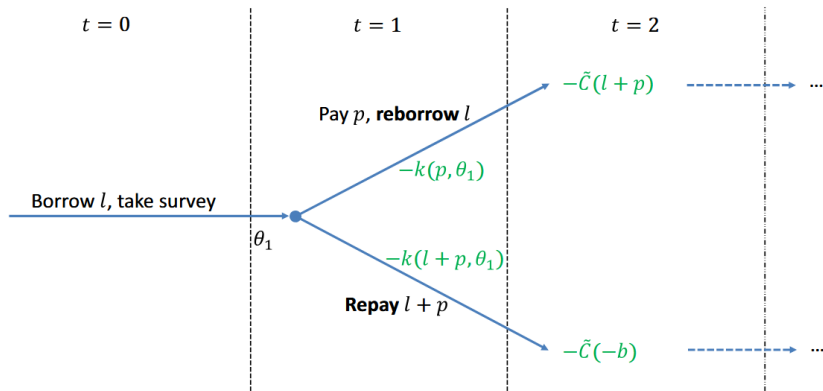
- $\mu(0)$: actual reborrowing probability in the control group
- $\tilde{\mu}(0)$: average predicted re-borrowing probability without the incentive
- $b = 100$: the amount of the no-borrowing incentive
- $\tilde{\Delta} := \tilde{\mu}(0) - \tilde{\mu}(b)$: predicted treatment effect of the incentive on borrowing

Let $b^\dagger$ be the distance of H and G; i.e., $\tilde{\mu}(b^\dagger) = \mu(0)$.

- A linear approximation yields

$$b^\dagger = \frac{b}{-\tilde{\Delta}}(\mu(0) - \tilde{\mu}(0)).$$

Timing of the Dynamic Game



Structural Estimation: Equal Demand

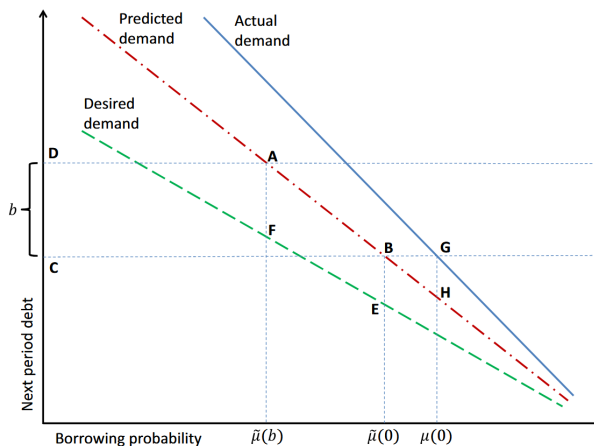
Equal demand: actual behavior vs. predicted behavior

$$\underbrace{\beta \left[\tilde{C}(l+p) - \tilde{C}(0) \right]}_{\text{period 1 self's reborrowing cost}} = \underbrace{\tilde{\beta} \left[\tilde{C}(l+p) - \tilde{C}(-b^\dagger) \right]}_{\text{predicted reborrowing cost with incentive } b^\dagger} .$$

Risk-neutrality implies linearity of C . Thus

$$\boxed{\frac{\beta}{\tilde{\beta}} = \frac{l+p+b^\dagger}{l+p}}$$

Identifying Perceived Present Focus



Define a time-consistent person's valuation of the incentive by

$$w^*(b) := b - ABCD = b - b \frac{\tilde{\mu}(0) + \tilde{\mu}(b)}{2},$$

equivalent to receiving b cash and facing a b price increase on reborrowing.

Identifying Perceived Present Focus (Cont.)

A time-inconsistent borrower's valuation of the incentive is

$$\begin{aligned} w(b) &:= b - ABCD + ABEF \\ &= w^*(b) + \underbrace{\left(1 - \tilde{\beta}\right)}_{\text{perceived internality}} \underbrace{\left(l + p + \frac{b}{2}\right)}_{\text{internality reduction benefit}} \underbrace{\tilde{\Delta}}_{\text{predicted behavior change}} \end{aligned}$$

They value the incentive more since it makes the future self borrow less.

- The per-unit-of-behavior-change value of this reduction is the vertical gap between the Desired and Predicted demand curves.
- Gap at x : desired cost - predicted cost $= (l + p + x) - \tilde{\beta}(l + p + x)$

Formal Model with Risk Aversion

Account for absolute risk aversion $\alpha := \frac{\tilde{C}''(0)}{\tilde{C}'(0)}$ at $x = 0$, and define $\rho := \alpha(l + p)$.

Proposition 1 (informal)

Omitting higher-order terms, we have

$$\underbrace{\beta(l + p) \left(1 + \frac{\rho}{2}\right)}_{\text{period 1 self's reborrowing cost}} = \underbrace{\tilde{\beta} \left(l + p + b^\dagger\right) \left(1 + \frac{\rho}{2} - \frac{\alpha}{2} b^\dagger\right)}_{\text{predicted reborrowing cost with incentive } b^\dagger},$$

and

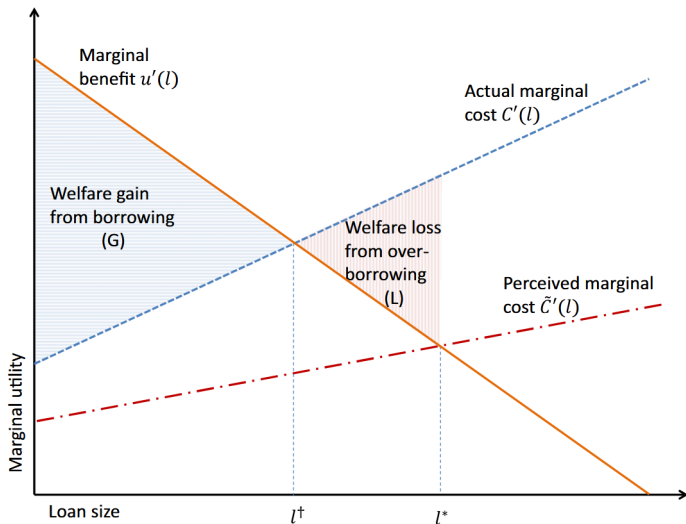
$$\begin{aligned} & \left(1 - \frac{\alpha w}{2}\right) \left\{ \underbrace{w \cdot \left(1 + \rho \left(\tilde{\mu}(0) + \frac{w\rho}{b} \frac{\tilde{\Delta}}{2}\right)\right)}_{\text{utility from } w \text{ if time consistent}} - \underbrace{(1 - \tilde{\beta}) \left(1 + \frac{\rho}{2}\right) (l + p)}_{\text{perceived internality}} \underbrace{\frac{w\rho}{b} \tilde{\Delta}}_{\text{behavior change}} \right\} \\ &= \left(1 - \frac{\alpha b}{2}\right) \left\{ \underbrace{b \cdot \left(1 - \tilde{\mu}(0) + \frac{\tilde{\Delta}}{2}\right)}_{\text{utility from } b \text{ if time consistent}} + \underbrace{(1 - \tilde{\beta}) \left(1 + \frac{\rho}{2}\right) \left(l + p + \frac{b}{2}\right)}_{\text{perceived internality}} \underbrace{\tilde{\Delta}}_{\text{behavior change}} \right\}. \end{aligned}$$

Naivete and Present Focus Parameters

TABLE 3
Partially naive present focus parameters

	α	(1) Estimated $\beta/\tilde{\beta}$	(2) Estimated $\tilde{\beta}$	(3) Estimated β
Full sample	0.0064	0.98 (0.95, 0.99)	0.76 (0.73, 0.78)	0.74 (0.71, 0.77)
Full sample	0.002	0.96 (0.93, 0.99)	0.77 (0.74, 0.79)	0.74 (0.70, 0.77)
Full sample	0.0005	0.95 (0.91, 1.00)	0.83 (0.80, 0.85)	0.79 (0.75, 0.83)
Full sample	0.0002	0.95 (0.90, 0.99)	0.85 (0.83, 0.88)	0.81 (0.77, 0.85)
Full sample	0	0.95 (0.90, 0.99)	0.87 (0.85, 0.90)	0.83 (0.78, 0.87)
0–3 loans in past six months	0.0064	0.89 (0.71, 0.94)	0.78 (0.74, 0.82)	0.70 (0.52, 0.75)
0–3 loans in past six months	0	0.79 (0.59, 0.91)	0.92 (0.87, 0.97)	0.73 (0.54, 0.85)
4+ loans in past six months	0.0064	1.00 (0.97, 1.01)	0.75 (0.72, 0.78)	0.75 (0.71, 0.77)
4+ loans in past six months	0	1.00 (0.95, 1.05)	0.86 (0.83, 0.89)	0.86 (0.80, 0.90)
Group by loans in past six months	0.0064	0.96 (0.90, 0.98)	0.76 (0.73, 0.78)	0.73 (0.68, 0.75)
Group by loans in past six months	0	0.94 (0.86, 0.98)	0.87 (0.85, 0.90)	0.82 (0.75, 0.86)

Welfare and Policy Implications: Graphical Analysis



Welfare and Policy Implications: Proposition 2

Borrowers receive a transitory shock θ_t and serially correlated shock η_t in each period $t \geq 1$.

Proposition 2

Suppose that the unconditional distribution of η is time invariant. Let μ be the empirically observed re-borrowing rate. Relative to the repayment costs $C^{TC}(I)$ of time-consistent borrowers, the repayment costs $C_{\beta}^S(I)$ of borrowers with present focus β and $\tilde{\beta}_1 = \beta$ are bounded by

$$C_{\beta}^S(I) \leq \frac{C^{TC}(I)}{\beta}.$$

If $\mu < 1$, the repayment costs $C_{\beta, \tilde{\beta}_1}^{PN}(I)$ of partially naive borrowers with present focus β and long-run beliefs $\tilde{\beta}_1 > \beta$ are bounded by

$$C_{\beta, \tilde{\beta}_1}^{PN}(I) \leq \frac{C^{TC}(I) - \mu\beta C_{\tilde{\beta}_1}^S(I)}{1 - \mu}.$$

Welfare and Policy Implications: Bounds

Inequalities bound the ratio of perceived to actual costs given $\beta, \tilde{\beta}, \mu$.

Sophisticates ($\beta_1 = \tilde{\beta}$):

- Present focus cannot increase costs by more than proportion $1/\beta$.

Partially and persistently naive ($\beta_1 > \tilde{\beta}$):

- Welfare losses only become large as $\mu \rightarrow 1$ (perpetual reborrowing).

Welfare and Policy Implications: Estimates

Plugging in the estimated parameters gives

- Present focus increases repayment costs by at most 29%.
- Temporary naivete leads borrowers to underestimate costs by at most 7% ($\tilde{C}/C = 0.93$).
- Persistent partial naivete makes them underestimate costs by at most 6% ($\tilde{C}/C = 0.94$).

Even if we assume $\tilde{C}/C = 0.9$, it is unlikely that Loss > Gain.

- Loss > Gain requires demand to be too elastic.

Numerical Simulation

Parametric setup

- $p(l)/l = 0.15$, 15% borrow rate
- $\delta = 0.998$ (5% annual discount rate given two-week pay cycle)
- $\alpha = 0.002$ or 0.0002
- $(\beta, \tilde{\beta})$ different specifications
- Other parameters calibrated using Lender's 2017 data

Assume a \$500 loan size cap at baseline.

Numerical Simulation: Results

Rollover restriction refers to borrowers repaying by period 3 at the latest.

TABLE 5
Borrower welfare under payday lending regulations

(a) Higher demand elasticity					
Row	Scenario	(1) Baseline (\$500 cap) (%)	(2) \$400 cap (%)	(3) Rollover restriction (%)	(4) 25% fee (%)
1	$\beta = 1, \tilde{\beta} = 1$	100.0	92.0	99.8	94.9
2	$\beta = 0.74, \tilde{\beta} = 0.77$ (primary estimates)	95.9	88.7	97.4	90.3
3	$\beta = 0.74, \tilde{\beta} = 0.77$, consume in $t=0$	95.2	88.3	96.8	89.4
4	Heterogeneous	90.8	84.4	97.3	84.4
5	$\beta = 0.74, \tilde{\beta} = 1$	91.1	84.8	97.1	84.5
6	$\beta = 0.74, \tilde{\beta} = 0.74$	96.7	89.3	97.4	91.2
7	$\beta = 0.74, \tilde{\beta}_0 = 0.88, \tilde{\beta}_1 = 0.74$ (learning)	95.1	88.0	97.2	88.8
8	Primary, heterogeneous, learning, consume in $t=0$	90.2	84.5	96.0	82.8
9	$\beta = 0.63, \tilde{\beta} = 0.86$ (expert forecast)	80.8	76.4	95.8	71.8
10	$\beta = 0.63, \tilde{\beta} = 0.86$, consume in $t=0$	77.0	74.3	94.3	66.2

(b) Lower demand elasticity					
Row	Scenario	(1) Baseline (\$500 cap) (%)	(2) \$400 cap (%)	(3) Rollover restriction (%)	(4) 25% fee (%)
1	$\beta = 1, \tilde{\beta} = 1$	100.0	89.3	99.5	86.1
2	$\beta = 0.74, \tilde{\beta} = 0.77$ (primary estimates)	88.8	80.4	92.8	74.6
3	$\beta = 0.74, \tilde{\beta} = 0.77$, consume in $t=0$	86.0	78.9	90.5	70.4

Numerical Simulation: Summary

Summary

- Even under worst-case specifications, baseline welfare is positive → borrowers benefit from credit access.
- Bans and tighter caps reduce welfare in every specification.
- Rollover restrictions increase welfare by contrast.

This presentation omits the following content:

- Expert survey from payday lending and behavioral economics experts.
- Technical details of borrowers' microfoundation and dynamic game
- Real-life regulatory measures on payday loans in the USA
- Several secondary results, discussions and literature reviews

Limitations

1. Borrowing spells are exogenous events with initial liquidity demand.
2. Learning is modeled simplistically; evidence is cross-sectional.
3. Potential externalities of high-interest loans are not accounted for.
4. Their results mainly apply to formal high-interest loans.

Takeaways

- High-interest loans are not always predatory.
 - Payday loan borrowers exhibit sophistication by almost correctly predicting their likelihood of reborrowing.
 - Payday loan borrowers perceive themselves to have present focus.
- Complete bans and tighter loan caps reduce borrowers' welfare, while rollover restrictions generally increase their welfare.
- This paper first incorporates stochastic repayment shocks formally, and shows the infinite debt trap predicted by deterministic models is unlikely the empirical outcome.